



**NATIONAL SKILL DEVELOPMENT
CORPORATION
In Collaboration with
YUVA INTERN**

Virtual R Data Analyst Intern

WEEK 3 REPORT

**Statistical Analysis and Predictive Modeling using
R**

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1. INTRODUCTION

The goal for Week 3 was to conduct a statistical analysis and build a forecasting model in R. In this regard, the supermarket sales dataset used previously was further analyzed. The analysis aimed at data distribution analysis, statistical correlation testing, forecasting model creation, verification of its effectiveness, and potential enhancements. The variable chosen for prediction was customer type, which had two subdivisions: Member and Normal.

2. EXPLORATORY STATISTICAL ANALYSIS

2.1 Descriptive Statistics

Descriptive statistics were performed to understand the characteristics of the numerical variables used in the analysis.

```
summary(
  model_data %>%
    select(unit_price, quantity, rating, month, day, hour)
)
```

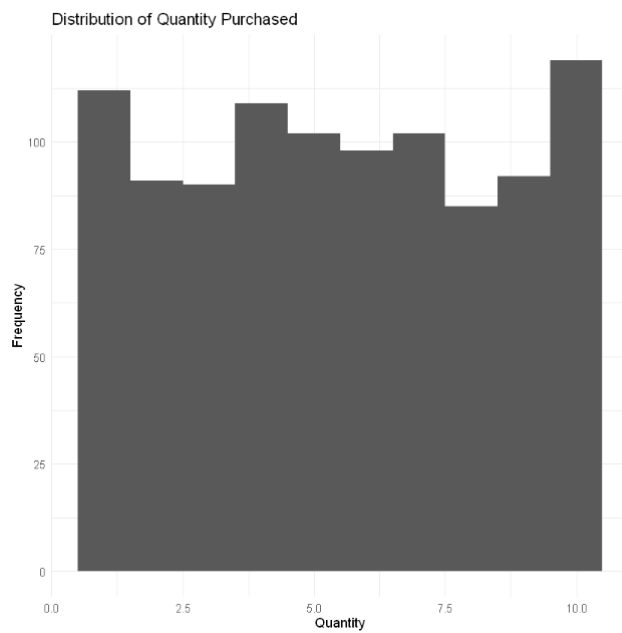
unit_price		quantity		rating		month	
Min.	:10.08	Min.	: 1.00	Min.	: 4.000	Jan	:352
1st Qu.	:32.88	1st Qu.	: 3.00	1st Qu.	: 5.500	Mar	:345
Median	:55.23	Median	: 5.00	Median	: 7.000	Feb	:303
Mean	:55.67	Mean	: 5.51	Mean	: 6.973	Apr	: 0
3rd Qu.	:77.94	3rd Qu.	: 8.00	3rd Qu.	: 8.500	May	: 0
Max.	:99.96	Max.	:10.00	Max.	:10.000	Jun	: 0
						(Other):	0

day		hour	
Length	:1000	Min.	:10.00
N.unique	: 7	1st Qu.	:12.00
N.blank	: 0	Median	:15.00
Min.nchar	: 6	Mean	:14.91
Max.nchar	: 9	3rd Qu.	:18.00
		Max.	:20.00

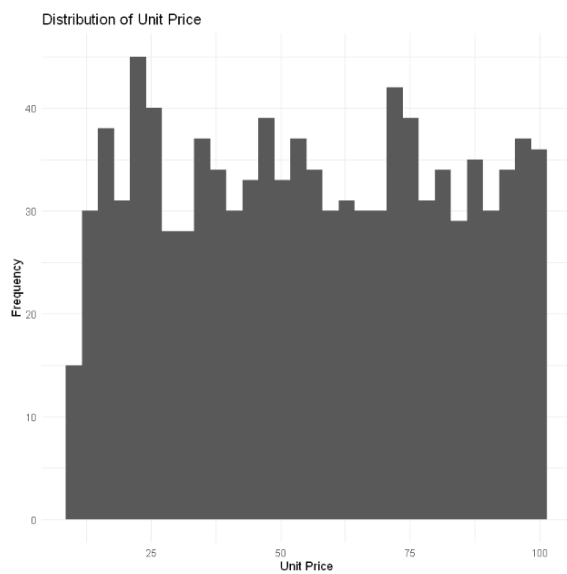
2.2 Distribution Analysis

Histograms were used to examine the distributions of important numerical variables.

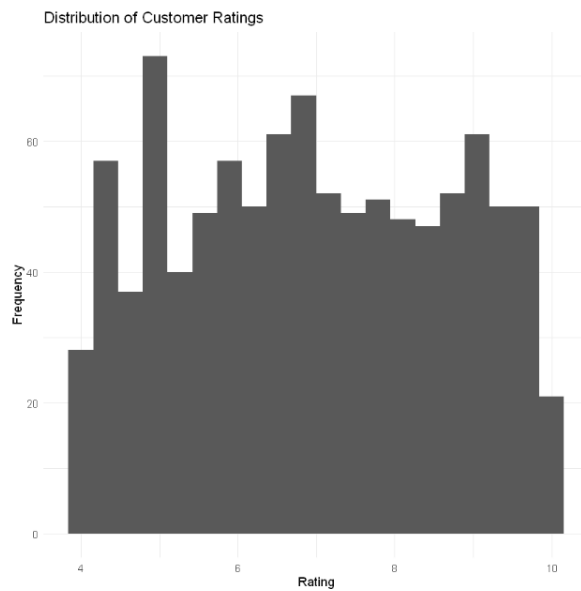
```
ggplot(model_data, aes(x = quantity)) +
  geom_histogram(bins = 10) +
  labs(title = "Distribution of Quantity Purchased",
    x = "Quantity", y = "Frequency") +
  theme_minimal()
```



```
ggplot(model_data, aes(x = unit_price)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Unit Price",
        x = "Unit Price", y = "Frequency") +
  theme_minimal()
```



```
ggplot(model_data, aes(x = rating)) +
  geom_histogram(bins = 20) +
  labs(title = "Distribution of Customer Ratings",
        x = "Rating", y = "Frequency") +
  theme_minimal()
```



2.3 Normality Testing

The Shapiro-Wilk test was used to examine whether the numerical variables followed a normal distribution.
 H_0 : The variable follows a normal distribution. H_1 : The variable does not follow a normal distribution.

```
shapiro.test(model_data$quantity)
shapiro.test(model_data$unit_price)
shapiro.test(model_data$rating)
```

Shapiro-Wilk normality test

```
data: model_data$quantity
W = 0.93258, p-value < 2.2e-16
```

Shapiro-Wilk normality test

```
data: model_data$unit_price
W = 0.95188, p-value < 2.2e-16
```

Shapiro-Wilk normality test

```
data: model_data$rating
W = 0.9582, p-value = 2.688e-16
```

2.4 Correlation Analysis

Correlation analysis was performed to examine relationships between numerical variables.

```
correlation_data <- model_data %>%
  select(where(is.numeric))
cor_matrix <- cor(correlation_data, use = "complete.obs")
cor_matrix

corrplot(cor_matrix, method = "number", type = "upper")
```

```
'unit_price' · 'quantity' · 'rating' · 'hour'
```

A matrix: 4 × 4 of type dbl

	unit_price	quantity	rating	hour
unit_price	1.0000000000	0.010777564	-0.008777507	0.008242210
quantity	0.010777564	1.0000000000	-0.015814905	-0.007316886
rating	-0.008777507	-0.015814905	1.0000000000	-0.030587644
hour	0.008242210	-0.007316886	-0.030587644	1.0000000000

3. HYPOTHESIS TESTING

3.1 T-Test: Customer Type and Quantity

H₀: There is no significant difference in mean quantity purchased between Member and Normal customers.

H₁: There is a significant difference in mean quantity purchased between Member and Normal customers.

```
t_test_quantity <- t.test(  
  quantity ~ customer_type,  
  data = model_data  
)  
t_test_quantity
```

Welch Two Sample t-test

```
data: quantity by customer_type  
t = 1.4867, df = 945.97, p-value = 0.1374  
alternative hypothesis: true difference in means between group Member and group Normal is not equal to 0  
95 percent confidence interval:  
-0.08833615 0.64046717  
sample estimates:  
mean in group Member mean in group Normal  
5.630088 5.354023
```

3.2 Chi-Square Test: Customer Type and Gender

H₀: Customer type and gender are independent.

H₁: Customer type and gender are associated.

```
gender_table <- table(model_data$customer_type, model_data$gender)  
chi_gender <- chisq.test(gender_table)  
chi_gender
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: gender_table  
X-squared = 17.962, df = 1, p-value = 2.253e-05
```

3.3 Chi-Square Test: Customer Type and Payment

H₀: Customer type and payment method are independent.

H₁: Customer type and payment method are associated.

```
payment_table <- table(model_data$customer_type, model_data$payment)
chi_payment <- chisq.test(payment_table)
chi_payment
```

Pearson's Chi-squared test

```
data: payment_table
X-squared = 2.6714, df = 2, p-value = 0.263
```

4. PREDICTIVE MODEL DEVELOPMENT

4.1 Model Selection

The reason logistic regression was chosen is that its dependent variable, `customer_type`, is a binary categorical variable having two values: Member and Normal. This model uses predictors like gender, product line, unit price, quantity, ratings, payment method, month, day of the week, and time, but it doesn't consider the variable `customer_type_encoded`, since it is capturing the same information as the dependent variable.

4.2 Train-Test Split

```
set.seed(123)
train_index <- sample(1:nrow(model_data), size = 0.80 * nrow(model_data))
train_data <- model_data[train_index, ]
test_data <- model_data[-train_index, ]
nrow(train_data)
nrow(test_data)
```

```
# Part 14 - Train/Test Split

set.seed(123)

# Create random training indices
train_index <- sample(
  1:nrow(model_data),
  size = 0.80 * nrow(model_data)
)

# Create training data
train_data <- model_data[train_index, ]

# Create testing data
test_data <- model_data[-train_index, ]

# Check sizes
nrow(train_data)
nrow(test_data)
```

800

200

4.3 Logistic Regression

```
logistic_model <- glm(
  customer_type ~ gender + product_line +
    unit_price + quantity + rating +
    payment + month + day + hour,
  data = train_data,
  family = binomial
)
summary(logistic_model)

Call:
glm(formula = customer_type ~ gender + product_line + unit_price +
    quantity + rating + payment + month + day + hour, family = binomial,
    data = train_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      0.194910   0.588276   0.331 0.740400
genderMale        0.558349   0.149418   3.737 0.000186 ***
product_lineFashion accessories  0.043168   0.249062   0.173 0.862399
product_lineFood and beverages -0.185596   0.247755  -0.749 0.453788
product_lineHealth and beauty  -0.034429   0.262456  -0.131 0.895632
product_lineHome and lifestyle -0.115349   0.261135  -0.442 0.658691
product_lineSports and travel  -0.113935   0.256704  -0.444 0.657160
unit_price        -0.004880   0.002787  -1.751 0.079995 .
quantity          -0.030832   0.025648  -1.202 0.229321
rating            0.022432   0.043357   0.517 0.604886
paymentCredit card -0.262909   0.183115  -1.436 0.151071
paymentWallet      0.116437   0.175484   0.664 0.506996
month.L           -0.005246   0.125449  -0.042 0.966642
month.Q           0.176089   0.129891   1.356 0.175203
dayMonday         0.235427   0.279232   0.843 0.399159
daySaturday       -0.009235   0.268073  -0.034 0.972519
daySunday         -0.133415   0.283963  -0.470 0.638474
dayThursday       0.089505   0.277847   0.322 0.747350
dayTuesday       -0.462194   0.276827  -1.670 0.094996 .
dayWednesday      0.005762   0.277451   0.021 0.983432
hour             -0.016773   0.023184  -0.723 0.469397
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1098.4  on 799  degrees of freedom
Residual deviance: 1061.2  on 779  degrees of freedom
AIC: 1103.2

Number of Fisher Scoring iterations: 4
```

5. MODEL EVALUATION

5.1 Prediction

```
predicted_probability <- predict(
  logistic_model, newdata = test_data, type = "response"
)
predicted_class <- ifelse(predicted_probability >= 0.5, "Normal", "Member")
predicted_class <- factor(predicted_class, levels = levels(test_data$customer_type))
```

5.2 Confusion Matrix

```
confusion_matrix <- table(
  Actual = test_data$customer_type,
  Predicted = predicted_class
)
confusion_matrix
```

	Predicted	
Actual	Member	Normal
Member	90	29
Normal	57	24

5.3 Performance Metrics

```
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)
TP <- confusion_matrix["Member", "Member"]
FN <- confusion_matrix["Member", "Normal"]
FP <- confusion_matrix["Normal", "Member"]
TN <- confusion_matrix["Normal", "Normal"]
sensitivity <- TP / (TP + FN)
specificity <- TN / (TN + FP)
```

Metric	Result
Accuracy	0.57
Sensitivity	0.7563025
Specificity	0.2962963

6. CROSS-VALIDATION

Ten-fold cross-validation was performed to evaluate the consistency of the logistic regression model across multiple subsets of the training data.

```
set.seed(123)
k <- 10
folds <- sample(rep(1:k, length.out = nrow(train_data)))
cv_accuracy <- numeric(k)

for (i in 1:k) {
  cv_train <- train_data[folds != i, ]
  cv_test <- train_data[folds == i, ]
  cv_model <- glm(
    customer_type ~ gender + product_line + unit_price +
      quantity + rating + payment + month + day + hour,
    data = cv_train, family = binomial
  )
  cv_probability <- predict(cv_model, newdata = cv_test, type = "response")
  cv_prediction <- ifelse(cv_probability >= 0.5, "Normal", "Member")
  cv_accuracy[i] <- mean(cv_prediction == cv_test$customer_type)
}
mean_cv_accuracy <- mean(cv_accuracy)
mean_cv_accuracy_percentage <- mean_cv_accuracy * 100
mean_cv_accuracy_percentage
```

```
[78]: # Part 23 - 10-Fold Cross-Validation
set.seed(123)
k <- 10
folds <- sample(
  rep(1:k, length.out = nrow(train_data))
)
cv_accuracy <- numeric(k)
for (i in 1:k) {
  cv_train <- train_data[folds != i, ]
  cv_test <- train_data[folds == i, ]
  cv_model <- glm(
    customer_type ~ gender +
    product_line +
    unit_price +
    quantity +
    rating +
    payment +
    month +
    day +
    hour,
    data = cv_train,
    family = binomial
  )
  cv_probability <- predict(
    cv_model,
    newdata = cv_test,
    type = "response"
  )
  cv_prediction <- ifelse(
    cv_probability >= 0.5,
    "Normal",
    "Nonnormal"
  )
  cv_accuracy[i] <- mean(
    cv_prediction == cv_test$customer_type
  )
}
cv_accuracy
0.4875 0.5875 0.55 0.6375 0.6125 0.6375 0.55 0.5875 0.5625 0.4625
```

```
[79]: mean_cv_accuracy <- mean(cv_accuracy)
```

```
mean_cv_accuracy
```

```
0.5675
```

```
[80]: mean_cv_accuracy_percentage <- mean_cv_accuracy * 100
```

```
mean_cv_accuracy_percentage
```

```
56.75
```

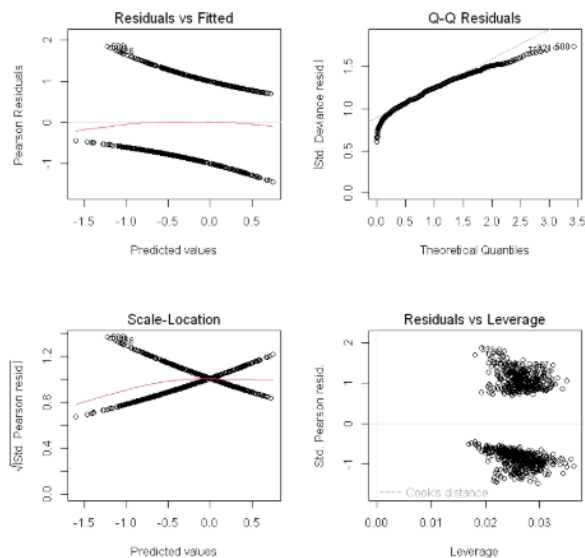
7. MODEL DIAGNOSTICS

Diagnostic analysis was performed to examine the fitted logistic regression model.

```
par(mfrow = c(2, 2))
```

```
plot(logistic_model)
```

```
par(mfrow = c(1, 1))
```



8. MODEL OPTIMIZATION

8.1 Improved Logistic Regression Model

A second logistic regression model was developed using a reduced set of predictors to determine whether a simpler model could maintain or improve classification performance.

```
improved_model <- glm(
  customer_type ~ gender + product_line +
    quantity + payment + rating,
  data = train_data,
  family = binomial
)
summary(improved_model)
```

Call:
glm(formula = customer_type ~ gender + product_line + quantity +
 payment + rating, family = binomial, data = train_data)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.341169	0.399728	-0.854	0.393381
genderMale	0.557039	0.146765	3.795	0.000147 ***
product_lineFashion accessories	0.001173	0.244044	0.005	0.996164
product_lineFood and beverages	-0.223640	0.243319	-0.919	0.358031
product_lineHealth and beauty	-0.057194	0.257104	-0.222	0.823960
product_lineHome and lifestyle	-0.109423	0.255059	-0.429	0.667914
product_lineSports and travel	-0.139653	0.251419	-0.555	0.578582
quantity	-0.031062	0.025306	-1.227	0.219658
paymentCredit card	-0.248527	0.180769	-1.375	0.169184
paymentEwallet	0.120662	0.173240	0.697	0.486114
rating	0.021845	0.042784	0.511	0.609633

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1098.4 on 799 degrees of freedom
Residual deviance: 1073.9 on 789 degrees of freedom
AIC: 1095.9

Number of Fisher Scoring iterations: 4

```
improved_probability <- predict(improved_model, newdata = test_data, type = "response")
improved_class <- ifelse(improved_probability >= 0.5, "Normal", "Member")
improved_class <- factor(improved_class, levels = levels(test_data$customer_type))
improved_confusion <- table(Actual = test_data$customer_type, Predicted = improved_class)
improved_accuracy <- sum(diag(improved_confusion)) / sum(improved_confusion)
improved_accuracy_percentage <- improved_accuracy * 100
improved_accuracy_percentage
```

```
improved_accuracy <- sum(
  diag(improved_confusion)
) / sum(improved_confusion)

improved_accuracy
```

0.58

```
improved_accuracy_percentage <- improved_accuracy * 100

improved_accuracy_percentage
```

58

9. MODEL PERFORMANCE AND INTERPRETATION

```
model_comparison <- data.frame(  
  Model = c("Full Logistic Regression", "Improved Logistic Regression"),  
  Accuracy = c(accuracy, improved_accuracy)  
)  
model_comparison$Accuracy_Percentage <- model_comparison$Accuracy * 100  
model_comparison
```

A data.frame: 2 × 3

Model	Accuracy	Accuracy_Percentage
<chr>	<dbl>	<dbl>
Full Logistic Regression	0.57	57
Improved Logistic Regression	0.58	58

The two models were compared based on classification accuracy. The model with the higher accuracy demonstrated better performance on the test dataset.

Potential Improvements:

- Testing additional classification algorithms.
- Performing more extensive feature selection.
- Tuning the classification threshold.
- Using a larger or more diverse dataset.
- Applying additional cross-validation and optimization techniques.

10. CONCLUSION

In Week 3, the focus was on statistical analysis and predictive modeling with R. The supermarket sales dataset underwent analysis through descriptive statistics, distribution, normality assessment, correlation tests, and hypothesis tests.

A logistic regression classification model was created to predict the type of customer. The data was divided into training and testing sets and evaluated using a confusion matrix, accuracy, sensitivity, specificity, and ten-fold cross-validation.

Model diagnostics were carried out, and an enhanced logistic regression model was created through a reduced set of predictors. The models were compared for predictive capabilities.

In conclusion, the task showed the use of statistical analysis and predictive modeling with R in performing analysis and drawing insights from transaction data.

11. APPENDIX

Additional R code, outputs, and screenshots can be included here if they are too large for the main sections of the report.